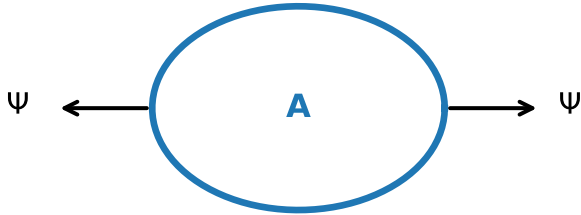


One-loop 1PI diagrams in CMSTG (SIM104)

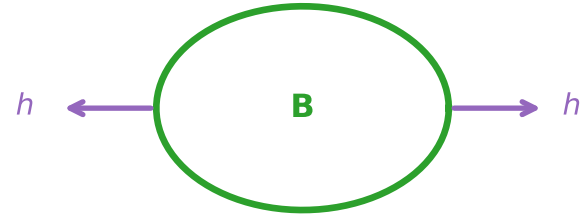


ψ self-energy

$$\Sigma_A = \Lambda_0^2 k_m^4 / (64\pi^2)$$

Bare: k_{UV}^4 (quartic)

$$\rightarrow \delta m^2/m_0^2 \approx 1 \text{ at } k_m^{\text{nat}}$$

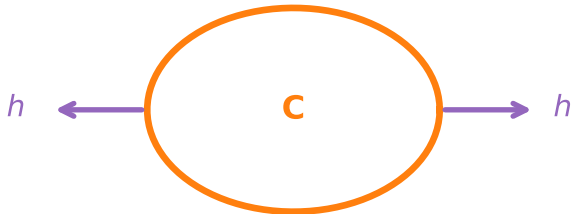


Graviton wfn renorm

$$\Pi_{hh}(0) = 0 \text{ (Ward identity)}$$

Protected, $dI/dp^2|_0$ finite

$$\rightarrow \Delta Z_h/Z_h \sim 10^{-6}$$

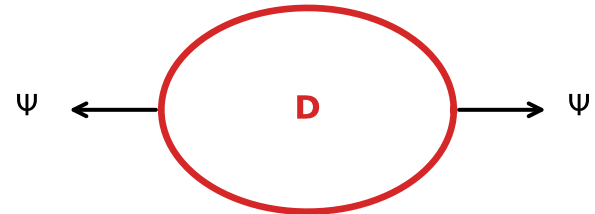


Vertex correction

$$\delta\Lambda_0/\Lambda_0 = \Lambda_0^2 k_m^2 / (16\pi^2 m_0^2)$$

Bare: k_{UV}^2 (quadratic)

$$\rightarrow \delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$$



Two-loop ψ self-energy

$$O(\Lambda_0^4)$$

$O(\Lambda_0^4)$ suppressed

$$\rightarrow \Sigma_D/\Sigma_A \approx 3.4 \times 10^{-6}$$